

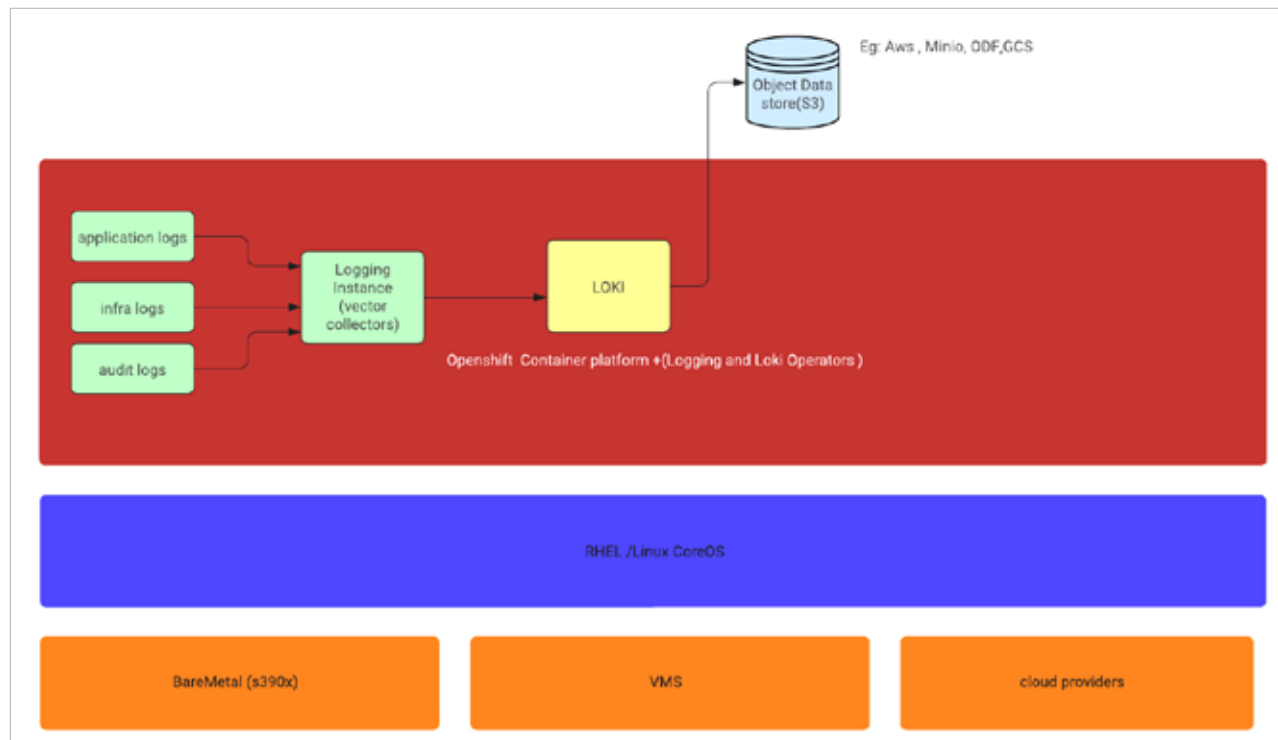


Figure 1: Architecture of the logging operator in s390x architecture

encourages transitioning to Vector as an alternative.

Log store: The log store serves as the repository for storing log data, facilitating analysis and retrieval. It is the default output destination for the log forwarder.

OpenShift provides options for the log store, including the default LokiStack log store or the legacy Elasticsearch log store. Additionally, logs can be forwarded to external log stores as needed.

The OpenShift Elasticsearch Operator is deprecated and will be phased out in future releases. Red Hat recommends using the Loki Operator as an alternative for managing the default log storage.

Visualisation: Visualisation components enable users to view log data in a graphical format, facilitating search, query, and analysis. The UI component, typically accessed through the OpenShift Container Platform web console, provides an intuitive interface for log exploration.

Steps to deploy logging on OpenShift

To start the installation process, access the web console and navigate to *Operators -> Operator Hub*. Figure 2 illustrates the available RedHat OpenShift logging operators in the Operator Hub.

Select 'Red Hat OpenShift Logging' from the list of operators provided and proceed by clicking 'Install' to initiate the installation.

While installing, ensure that the recommended namespace for the operator is 'openshift-logging' within the installed namespace.

Additionally, set the update channel for the operator to 'Stable'. You have the option to choose between the 'Automatic' or 'Manual' approval strategy.

With the automatic option, OLM (operator lifecycle manager) will handle updates automatically when newer versions are available. Alternatively, manual approval requires user intervention for updates. Figure 3 offers a preview of

the operator with the *Install* option.

Click on 'Install' to commence the installation process. Figure 4 displays all the necessary channels for installation.

After installation, verify the successful installation of the logging operator. Figure 5 shows the installation process is successfully completed.

Types of logs

Logging within the OpenShift Container Platform categorises logs into three main types.

Application logs: Generated by user applications running within the cluster, excluding infrastructure container applications.

Infrastructure logs: Include container logs from infrastructure namespaces such as OpenShift, kube, or default, as well as journald messages from nodes.

Audit logs: These are logs generated by auditd, the node audit system, along with logs from services such as kube-apiserver, openshift-apiserver, and ovn project if enabled.